Yusuf Unlu

SENIOR SOFTWARE ENGINEER

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Summary

Senior Software Engineer with 14 years of experience in backend and full-stack development using Java, Spring, Python, JavaScript, and Linux. Specialized in scalable systems for finance, crypto, and media. Skilled in CI/CD, microservices, event-driven design, and big data tools like Spark and Hadoop. Proficient in real-time data handling, WebSocket streaming, and cloud infrastructure with AWS, Docker, and Terraform. Holds two master's degrees and has experience teaching software development with Python. Maintains open-source projects, including a Java trading advisor.

Experience

Bank of America, Senior Software Engineer — July 2023-Present, NJ, USA

- Contributed to the **Self-Service Data Pipeline Builder**, an internal orchestration platform enabling users to visually design, execute, and manage data workflows using **Java** and **Spring**, executed on **Apache Spark** over **Hadoop**.
- Fixed **batch processing inefficiency** in the **Impala JDBC driver**, resulting in up to **100x performance improvement** during large-scale data ingestion; also configured **Hive** integration and resolved **charset issues** in **Teradata JDBC** to ensure reliable platform integrity.
- Managed and operated large-scale data workflows across **Hive**, **PostgreSQL**, **Spark/HDFS**, **Oracle**, **Teradata**, and **Impala**, focusing on charset handling and batch processing reliability.
- Collaborated with analysts, users, QA, and the development team to implement features and resolve bugs.
- Assisted backend developers with **JavaScript/Angular UI setup** on local environments and created onboarding and configuration documentation on **Confluence** for developers and stakeholders to improve collaboration.
- Wrote unit tests and monitored **SonarQube** coverage and vulnerabilities, proactively fixing issues to ensure code quality and security compliance.
- Led production deployment efforts including **SQL schema scripts**, **Ansible automation scripts** and **Jenkins builds**.
- **Tech Stack:** Java, Spring, Apache Spark, Hadoop, Parquet, CSV, Iceberg, Oracle, Teradata, Hive, Impala, FTP, HDFS, PostgreSQL, SQL, WebSocket, Ansible, Jenkins, JDBC, Spark APIs, UDFs, Angular, JavaScript, Confluence

Morgan Stanley, Senior Java Developer — April 2022-May 2023, NYC, USA

- Contributed to building a **highly scalable, resilient, event-driven system** for real-time client balance calculations (e.g., spending power, cash available), replacing a legacy mainframe solution. The system processes **8M+ daily calls**, supports **12M+ accounts**, and is used by **30+ business-critical applications**.
- Contributed to backend microservices development using **Java**, **Spring Boot**, **Kafka**, **MS SQL**, **Liquibase**, **MongoDB**, and **Avro** in a microservice architecture, with unit tests ensuring reliability and maintainability.
- Enhanced CI/CD workflows with **Jenkins**, **Gradle**, **Python**, **Docker**, and **Cloudify**, reducing manual effort and accelerating delivery cycles.
- Built an **MSSQL instance comparator** to support database deployments, minimizing downtime and increasing release reliability; also supported the implementation of a **blue-green deployment setup** with dual production environments and seamless traffic handoff.

Bitlo, Lead Software Engineer — May 2021-April 2022, Istanbul, Turkey

- Improved market maker bots and Order Management Services by integrating real-time **ticker data**, **order book updates**, and trade execution events using **WebSocket** and **STOMP** protocols at **Bitlo, a crypto market startup**.
- Led the CRM team and implemented it from scratch using **Java**, **Spring**, **ReactJS**, **Docker**, and **AWS** (EC2, RDS, ECR), developing **REST APIs** and features for **KYC/AML compliance** and **money transaction monitoring**.
- Managed setup and configuration of **Solana** and **Dogecoin** nodes for blockchain integration.
- Maintained and supported main project DevOps using **GitLab Pipeline**, **Terraform**, and **AWS**.

Deal Room Events, Senior Software Engineer — Sep 2020-April 2021, Espoo, Finland

- Backend development and CI/CD for Deal Room online conference system startup using Java, Firebase, **AWS Serverless** (Lambda, Kinesis, SQS, SNS, S3, DynamoDB, CodePipeline, CloudFront, CloudWatch, Cognito, API Gateway) in event sourcing architecture
- Built stream-publish project from scratch using FFmpeg, Bash, CloudFormation, **Terraform**, **Docker**, Groovy, Video.js, Python, RTMP, AWS (ECS, Fargate, ECR, EC2, AWS Java SDK)

Artiwise, **Senior Software Engineer** Feb 2020-July 2020, Istanbul, Turkey

- At **Artiwise,** customer Experience Management startup, implemented a Java/Spring application providing APIs that allow customers to define filters and retrieve customized data.
- Designed and developed scheduled processes to fuse and filter crawled data based on customer-specific criteria before pushing it to AI services for further analysis.
- Built data extraction and filtering logic using **Java**, **Hibernate ORM**, **Spring**, **PostgreSQL**, **Solr** (NoSQL), and **Redis**.
- Utilized Kafka for streaming and message-driven communication.
- Implemented centralized logging and monitoring with the **ELK** stack (**Elasticsearch**, **Logstash**, **Kibana**).
- Containerized and deployed the solution using **Docker** and **Kubernetes**.

Mobven, **Full Stack Developer** Aug 2018-Feb 2020, Istanbul, Turkey

- Developed Momentum Cloud Mobile Test Automation System using Spring Boot, PostgreSQL, ReactJS. Conducted device testing with Appium and adb scripts. Implemented video streaming via RTMP, FFmpeg, ActionScript and Nginx.
- Contributed to Mobven Mobile Payment MVP using Spring, PostgreSQL, Android, and various hashing/encryption algorithms for data over sound.
- Contributed Turkishairlines.com, using Spring, Eureka, OpenShift, Zuul, Knockout.js, CSS in microservice architecture.

Bein Media, **Software Engineer** April 2016-July 2018, Istanbul, Turkey

- I worked on Spring Boot, Android, Java, SQLite, Chromecast, HLS, Widevine, Fairplay, MPEG-DASH, DRM technologies.
- Contributed to Android and middleware of Mobile CRM app with over 200 screens.
- Implemented video player in Android streaming platform using ExoPlayer.

Vestel, **C, C++ Developer** July 2011-November 2014, Izmir, Turkey

- Developed cross-platform Linux kernel applications for embedded DVB devices using C, C++.
- Improved nightly and commit builds with Python and Bash scripts.
- Developed and integrated Android TV drivers working with Novatek platform engineers.
- Strong knowledge of Linux OS and layered architecture.

Education

Computer Science Master's Degree from CUNY, NY, USA
Computer Eng. Master's at Ege Uni. Izmir, Turkey (Not completed)
Electronics Engineering Bachelor's Degree from Ege Uni. Izmir, Turkey

Certificates

Hadoop Platform and Application Framework from UC San Diego in 2019
Spring Security from LinkedIn in 2021
React.js Essential from LinkedIn in 2021
Golang Complete Developer's Guide, Udemy
Video and Image Compression, Ozyegin University in 2013